



IIT Gandhinagar

PG Diploma in AI-ML & Agentic AI Engineering

Curriculum and Learning Journey





Credit Structure (40 Credits)

Component	Credits	Hours
Theory (Lectures)	17	170 hrs
Practical (Labs + Capstone)	16	320 hrs
Competitive Coding & Ideas Lab	7	140 hrs
TOTAL	40	630 hrs

Credit Calculation: 10 teaching hours = 1 credit OR 20 practical hours = 1 credit

20-Week Learning Journey

Phase	Timeline	Subjects & Topics	Job Skills Unlocked
Foundation	Weeks 1-6	Python for ML, Statistics, DSA, ML Fundamentals, Neural Networks	ML algorithms, model training, neural network basics, Python ML stack
Specialization	Weeks 7-14	Deep Learning & Vision, NLP & Transformers, Agentic AI, Multi-Agent Systems, MLOps	CNNs, BERT/GPT fine-tuning, RAG systems, AI agents, LLM orchestration
Midterm	Week 15	Comprehensive technical assessment	Mid-program evaluation, interview readiness check
Capstone	Weeks 16-19	Autonomous AI Agent System: Multi-agent collaboration, LLM orchestration, production deployment	Agent architecture, production AI systems, model monitoring, A/B testing
Final Exam	Between Week 15-20	ML interviews, system design, research paper discussions	Interview-ready for ML Engineer, Agentic AI Engineer, MLOps roles



Month-by-Month: Subjects, Tools & Job Skills

Month 1 (Weeks 1-6): Engineering Toolkit

- Subjects: Python for ML (6cr), Statistics (5cr), ML Fundamentals (6cr), Neural Networks (4cr)
- Tools: Python, PyTorch, TensorFlow, NumPy, scikit-learn, Jupyter, Git, Weights & Biases
- Job Skills: ML algorithms (regression, classification, trees), neural network training, backpropagation, optimization

Month 2 (Weeks 7-10): AI Specialization

- Subjects: Deep Learning & Vision (7cr), NLP & Transformers (7cr), Agentic AI (5cr), MLOps (3cr)
- Tools: Transformers (HuggingFace), LangChain, LlamaIndex, LangGraph, OpenAI/Claude APIs, Pinecone, ChromaDB, MLflow, Docker
- Job Skills: CNNs, object detection, BERT/GPT fine-tuning, RAG systems, AI agents, tool use, multi-agent orchestration

Months 3-4 (Weeks 11-17): Capstone Project

- Capstone (9cr): Autonomous AI agent system with planning, reasoning, tool use, memory, multi-agent collaboration
- Tools: Complete AI stack, LangGraph, vector databases, FastAPI, monitoring, experiment tracking (MLflow, W&B)
- Job Skills: Agent architecture, LLM orchestration, production AI deployment, model monitoring, A/B testing for AI

Month 5 (Weeks 18-20): Interview Mastery

- Focus: 300+ ML coding problems, implement algorithms from scratch, system design for ML, 15+ mock interviews, research papers
- Tools: LeetCode, ML interview platforms, paper implementation challenges, Kaggle
- Job Skills: ML theory mastery, coding neural networks from scratch, system design (recommendation, search), behavioral prep

Throughout 20 Weeks: Competitive Coding & Ideas Lab (7cr)

- Daily ML coding, implement papers, Kaggle competitions, AI ethics, research discussions, hackathons



Extra Value Pack

AI/ML Tools Mastery

- PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, Weights & Biases, MLflow, experiment tracking

GenAI & Agentic AI Stack

- LangChain, LlamaIndex, LangGraph, CrewAI, OpenAI/Anthropic APIs, vector databases (Pinecone, Weaviate, ChromaDB), 150+ AI prompts

MLOps & Production AI

- Docker, Kubernetes, AWS SageMaker, model serving, monitoring, A/B testing, CI/CD for ML, model versioning

Agile & Scrum for AI Projects

- ML experiment tracking, sprint planning for research, Jira for AI projects, model performance retrospectives

Professional Development

- 2 clubs: Entrepreneurs Club + AI Innovation Club (optional, Mon/Wed/Fri evenings). ML tournaments & research paper discussions



Daily Learning Structure

Time (24-hour)	Activity
07:45 - 09:15	Breakfast at IIT-GN Campus
09:15 - 10:45	Session 1A - Core Concepts: Instructor live-builds ML/AI systems (neural networks, transformers, agents) with real-time Q&A
11:00 - 12:45	Session 1B - Hands-On Labs: Model training, agent building, fine-tuning. Pair programming, live debugging, instructor support
12:45 - 14:00	Lunch at IIT-GN Campus + Squad Sync: Discuss model results, share wins, plan afternoon sprint
14:00 - 15:15	Session 2A - Production Patterns: Real-world ML systems, optimization techniques, research paper implementation, industry cases
15:30 - 17:00	Session 2B - Advanced Implementation: Agent orchestration, production AI systems, mini-projects, peer code reviews
17:00 - 17:30	Break + Snacks at IIT-GN Campus
17:30 - 18:00	Daily Standup: What shipped? What broke? Blockers? 2-min demos. Commit to GitHub
18:00 - 19:30	ML Coding Hour: Implement algorithms from scratch, solve ML problems, Kaggle practice, daily leaderboard
19:30 onwards	Dinner at IIT-GN Campus (until 20:45), then hostel
Saturday	Weekly assessment + model evaluation showcase + mock ML interviews + paper discussions + club activities (optional)

Note: All meals (breakfast, lunch, snacks, dinner) are at IIT Gandhinagar campus mess